

Mehul V. Warade

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Summary

I am currently pursuing Doctorate (PhD) in green distributed computing at Deakin University with extensive experience in research, teamwork, and project management. Project experiences include web/application development, backend server management and full stack development. Academic experience includes database management, IoT, software development and networking. Demonstrated excellent leadership and teamwork skills with ability to develop strong relationships. Recipient of the Global Citizenship Award at Deakin University which is awarded to people who have demonstrated excellent people skills, communication skills, working in teams globally, etc.

Work Experience Summary

January 2025 – present

Tutor: Digital Systems and Networking – EDT703

Deakin University, Melbourne.

- Enabling students to have hands on experience on how electronics helps in digital systems. Creating content for the unit. Marking student submissions.

July 2024 – November 2024

Technology/Tools: Linux, Wireshark, IoT, Resistors, Transistors

Tutor: Data and Information Management – EDT702

Deakin University, Melbourne.

- Managing and creating the unit content. Teaching students about database management and how to manage the data stored using structured query languages.

May 2023 – June 2024

Technology/Tools: MySQL, Oracle, JavaScript, React, REST, Linux

Lecturer: Advanced Embedded Systems – SIT329

Deakin University, Melbourne.

- Development of a IoT project with other colleagues. Implementing safety regulations. Providing students with support in developing their projects.

May 2023 – November 2023

Technology/Tools: Embedded Systems, Linux, Python, JS, HTML

Support Tutor: Concurrent and Distributed Programming – SIT315

Deakin University, Melbourne.

- Provide workshop and marking support for students practical development and tasks.

May 2023 – November 2023

Technology/Tools: Arduino, C#, Parallel programming, Python

Lecturer: Information Technology Systems and Innovation - SIT224

Deakin University, Melbourne.

- Teaching classes and tutorials. Marking student submission and developing unit content.

May 2023 – November 2023

Technology/Tools: Visualization tools, Miro, Lucidchart, Jira, Canva

Tutor: Data Capture Technologies - SIT123

Deakin University, Melbourne.

- Conducting tutorials and workshops for capturing and analyzing data/ development using IoT sensor/ devices.

March 2023 - July 2023

Technology/Tools: PowerBI, MySQL, PostgreSQL, C++

Lecturer: Embedded Systems Development - SIT210

June 2022 – October 2023

Deakin University, Melbourne.

- Undertaking IoT development. Conducting tutorials and workshops for practical development using IoT sensor/ devices.

Technology/Tools: Linux, Python, JS, HTML, CSS, C#

Tutor: Database Management - SIT102/ SIT 772

Deakin University, Melbourne.

- Teaching study material and marking for workshop tutorials. Creating content for the unit relating to Database Management, Retrieval and Modelling.

Technology/Tools: MySQL, Oracle, JavaScript, React, REST, Linux.

Research Assistant

Deakin University, Melbourne.

- Conducted research in a green-house approach towards modern High-Performance Computing (HPC) using low-powered ARM based Single Board Computers.

Technology/Tools: JavaScript, Linux, GIT (source control), PHP, REST APIs, Docker, MySQL.

October 2020 – February 2021

Skills

- Web Development (**CSS, HTML, NodeJS, React, Angular and Php**).
- Strong in software programming languages such as **C++, C#, Java, Android Programming, MySQL database, Batch Programming, JavaScript and Python**.
- Tools/ Frameworks – **Visual Studio/ Code, XCode, Unity, Docker, SQL, Yii2 and GIT**.
- DevOps (**Ansible and Vagrant**).
- Proficient in remote server management of **Amazon Web Services** and game development in **Unity**.
- Strong base working with **embedded systems and clusters management**. Cluster development, computation and management using **Pegasus, Condor, MPI and Montage**.
- Management and development of software for embedded systems such as Raspberry-pi, Arduino, Photon, Odroid and Panda-Board.
- Strong programming, debugging and analyzing skills.

Education and Training

July 2021 – Current

Doctor of Philosophy (PhD)

Deakin University, Melbourne

- Currently completing my doctorate with full scholarship and stipend with the focus on Artificial Intelligence and Cluster Computing.

June 2020 – July 2021

Honours in Software Engineering

Deakin University, Melbourne

- Extensive research conducted with focus on developing a framework for evaluation of parallel algorithms on distributed systems.
- Published a thesis and my first academic paper based on all the research conducted during the year.
- Received second highest score in the class with 92% for the research and thesis.

June 2017 – June 2020

Bachelor's in software engineering

Deakin University, Melbourne

- Achieved High Distinction (Average WAM: 88%).
- Worked in teams to develop software and web applications. Focused on remote management in AWS with extensive knowledge in DevOps and Full stack Development.

Course Projects and Internships

- **Honours Research:** Developed a platform-independent flexible framework for evaluation of different cluster configurations. The main focus was to analyze their performance as compared to their energy consumption.
- **IT Placement:** Developed an automated data collection and analysis software using python for marketing and research purposes. Also, worked with state-of-the-art embedded systems used in agricultural research. Researched and developed a web-software for visualization and analysis of big data collected by the robot.
- **IOT study Tour and Internship in Vietnam):** Collaborated with FPT University in Da Nang, Vietnam to develop a smart remote-controlled Arduino powered car with the obstacle detection system. Also, worked on Day Leave Management System for GMO Z. Com VN Labs in Vietnam as a DevOps Engineer.
- **Game Development in China:** Collaborated with students from Inner Mongolia University, China in developing a working arcade game in 2 weeks as a study abroad program. The game was presented, and our team "Shoreso Plan" received an HD grade in the study tour.
- **Developing Internet of Things:** Developed a fully functional robot which is controlled by a web page. This was developed as a group project with a group of three people. Secured HD in the unit.

Research Paper and Publications

1. Mehul Warade, Jean-Guy Schneider, and Kevin Lee. 2021. FEPAC: A Framework for Evaluating Parallel Algorithms on Cluster Architectures. In 2021 Australasian Computer Science Week Multiconference (ACSW '21), February 1–5, 2021, Dunedin, New Zealand. ACM, New York, NY.
<https://dl.acm.org/doi/abs/10.1145/3437378.3444363>
2. Warade, Mehul, Jean-Guy Schneider, and Kevin Lee. "Measuring the Energy and Performance of Scientific Workflows on Low-Power Clusters." *Electronics* 11, no. 11 (2022): 1801.
<https://doi.org/10.3390/electronics11111801>
3. Warade, M., Schneider, J.G. and Lee, K., 2022, October. Towards Energy-aware Scheduling of Scientific Workflows. In 2022 International Conference on Green Energy, Computing and Sustainable Technology (GECOST) (pp. 93-98). IEEE.
<https://doi.org/10.1109/GECOST55694.2022.10010634>
4. Warade, M., Lee, K., Ranaweera, C. and Schneider, J.G., 2022, December. Energy Aware Adaptive Scheduling of Workflows. In 2022 IEEE Intl Conf on Parallel & Distributed Processing with Applications, Big Data & Cloud Computing, Sustainable Computing & Communications, Social Computing & Networking (ISPA/BDCloud/SocialCom/SustainCom) (pp. 562-570). IEEE.
<https://doi.org/10.1109/ISPA-BDCloud-SocialCom-SustainCom57177.2022.00078>
5. Warade, M., Lee, K., Ranaweera, C. and Schneider, J.G., 2023, June. Monitoring the Energy Consumption of Docker Containers. In 2023 IEEE 47th Annual Computers, Software, and Applications Conference (COMPSAC) (pp. 1703-1710). IEEE.
<https://doi.org/10.1109/COMPSAC57700.2023.00263>

6. Warade, M., Lee, K., Ranaweera, C. and Schneider, J.G., 2023, May. Optimising workflow execution for energy consumption and performance. In *2023 IEEE/ACM 7th International Workshop on Green And Sustainable Software (GREENS)* (pp. 24-29). IEEE.
<https://doi.org/10.1109/GREENS59328.2023.00011>
7. Warade, M., Lee, K., Ranaweera, C., & Schneider, J. G. (2024). Energy and Scientific Workflows: Smart Scheduling and Execution. *Journal of Information Science and Engineering*, 40, 957-977.
[https://doi.org/10.6688/JISE.202409_40\(5\).0002](https://doi.org/10.6688/JISE.202409_40(5).0002)
8. A Multi-User Energy-Aware Scheduler for Scientific Workflows. Submitted TPDS 2025
9. Energy-Aware Optimization and Scheduling of Scientific Workflows. Submitted ASE 2025

Referees

- Kevin Lee (Associate Profession – PhD Supervisor):
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- Jean-Guy Schneider (Professor - PhD Supervisor)
 - Email: jean-guy.schneider@monash.edu